PS2 Controller Roaster Control System

User Manual

ESP32-S3 N16R8 + PS2 Controller + TFT LCD + BLE Recipe Configuration

Firmware Version: DevKitC-1 v3.4.4 (PS2-Neutral-Fix + Recipe + NVS + Threshold NVS + PS2 Combo)

Supported Hardware

- ESP32-S3 N16R8 development board
- PS2 controller and receiver module
- ST7789 TFT LCD, 240 × 320 pixels
- Smartphone with BLE support
- Serial Bluetooth Terminal application
- Supports Artisan and HiBean

Chapter 1. System Overview

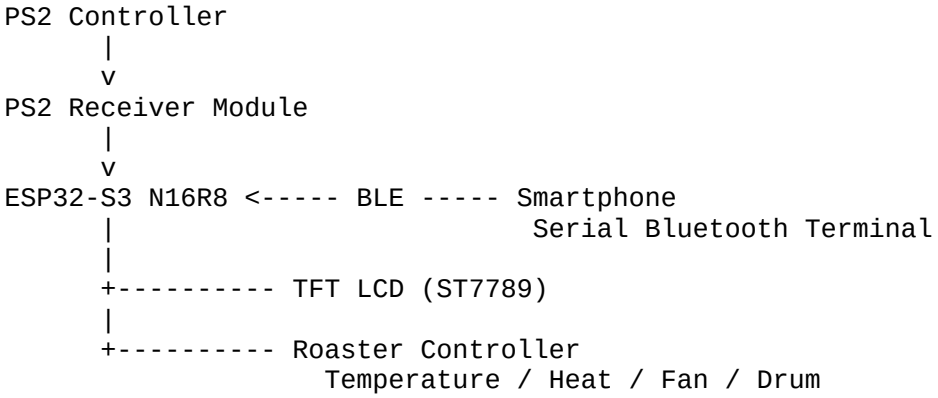
1.1 Main Features

This system uses an ESP32-S3 N16R8 as the main controller. A PS2 game controller provides manual operation, while a TFT LCD displays roasting information, temperatures, and curves.

Main features include:

1. Preheating start and cancellation.
2. Bean charging and bean dropping control.
3. Drum start and stop control.
4. Manual heat and airflow adjustment.
5. Roasting recipe selection.
6. Charge temperature adjustment.
7. Yellowing point (YEL) and First Crack (FC) temperature threshold adjustment.
8. Switching between text and curve display modes.
9. Editing roasting recipes through BLE.
10. Saving recipes and YEL/FC thresholds to non-volatile storage (NVS).

1.2 System Architecture



The actual heat, airflow, drum, and temperature control behavior depends on the complete roasting machine hardware and its controller configuration.

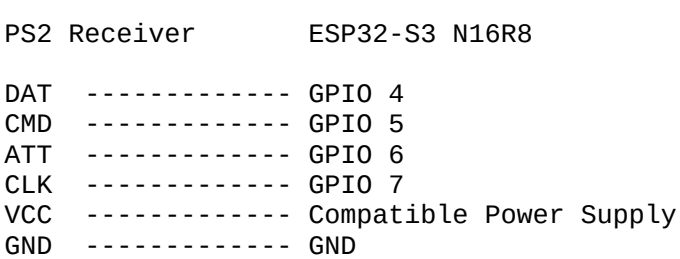
Chapter 2. Hardware Wiring

2.1 PS2 Receiver Module Wiring

Connect the PS2 receiver module to the ESP32-S3 according to the following GPIO configuration.

PS2 Receiver Pin	ESP32-S3 GPIO	Function
DAT / DATA	GPIO 4	Controller data input
CMD / COMMAND	GPIO 5	Controller command output
ATT / SEL / CS	GPIO 6	Controller select signal
CLK / CLOCK	GPIO 7	Communication clock
VCC	According to module specifications	Power supply
GND	GND	Common ground

Wiring diagram:



Important Notes

- The PS2 receiver and ESP32-S3 must share a common ground.
- Confirm the receiver module's required supply voltage before connecting power.
- If the receiver outputs 5 V logic signals, use an appropriate level shifter.
- PS2 pressure sensing and vibration functions are not enabled in this firmware.

- Controller initialization is attempted up to 10 times. If initialization fails, check the wiring and power supply first.

2.2 TFT LCD Wiring

This system uses an ST7789 TFT LCD with a resolution of 240 × 320 pixels and an SPI interface.

TFT LCD Pin	ESP32-S3 GPIO	Function
SCLK / SCL	GPIO 21	SPI clock
MOSI / SDA	GPIO 47	SPI data output
RST / RESET	GPIO 45	Display reset
DC / A0	GPIO 40	Data/command selection
CS	GPIO 41	Chip select
BLK / BL	GPIO 42	Backlight control
GND	GND	Common ground
VCC	According to module specifications	Power supply
MISO / SDO	Not connected	Configured as -1; unused

Wiring diagram:

```
TFT LCD (ST7789)     ESP32-S3 N16R8

SCLK  ------------- GPIO 21
MOSI  ------------- GPIO 47
RST   ------------- GPIO 45
DC    ------------- GPIO 40
CS    ------------- GPIO 41
BLK   ------------- GPIO 42
GND   ------------- GND
VCC   ------------- Compatible Power Supply
```

2.3 ESP32-S3 GPIO Reference Table

Function	GPIO
PS2 DAT	4
PS2 CMD	5
PS2 ATT	6
PS2 CLK	7
Drum PWM	9
TFT SCLK	21
TFT MOSI	47
TFT RST	45
TFT DC	40
TFT CS	41
TFT BLK	42
BLE / Other communication pins	According to firmware and board configuration

The firmware also uses UART GPIO 19 and 20, as well as LED GPIO 48. Do not reassign these pins without checking the complete hardware configuration.

Physical Pinout Reference

ESP32-S3 N16R8 development boards are available in different layouts. Always verify the GPIO labels against the actual board and its schematic. Use the GPIO numbers in this manual rather than relying solely on the physical position of the header pins.

2.4 Power and Electrical Safety

1. ESP32-S3 GPIO pins use 3.3 V logic and must not receive 5 V signals directly.
2. All modules must share a common ground.
3. Confirm the required supply voltage of the TFT LCD and PS2 receiver from their respective specifications.
4. Disconnect power before wiring or changing connections.
5. For initial testing, do not activate the heater or mechanical loads until the control signals have been verified.

Chapter 3. PS2 Controller Operation

3.1 Main Buttons

Button	Function	Operation
△ Triangle	Preheating control	Starts preheating; cancels during preheating or holding; toggles automatic drop during roasting
○ Circle	Charge beans	Executes the bean charging procedure
× Cross	Drop beans	Executes bean dropping or other state-dependent cooling/cleaning procedures
□ Square	Automatic drop	Toggles automatic bean dropping on/off during roasting
START	Drum control	Toggles the drum between running and stopped
L1	**Display mode**	**Switches between text and curve display modes**
SELECT	Threshold adjustment mode	Long-press to enter YEL/FC threshold adjustment mode
R3	Heat reset	Sets manual heat to 0
L3	Airflow reset	Sets airflow to 0

Important: Display switching has been reassigned to L1. SELECT is now used for YEL/FC threshold adjustment.

3.2 Directional Pad

During normal operation:

Button	Function
↑ Up	Select the next recipe
↓ Down	Select the previous recipe

← Left Decrease charge temperature by 5°C
→ Right Increase charge temperature by 5°C
The default charge temperature is 190°C, with a step size of 5°C. The manual adjustment range is 160–200°C.

The recipe menu contains a manual/default option and recipe entries. The available names and selections depend on the recipe data configured in the firmware.

3.3 Analog Sticks

Stick	Direction	Function
Right stick	Up / Down	Adjust heat
Left stick	Up / Down	Adjust airflow

Heat and airflow are adjusted in increments of 5 percentage points, within a range of 0–100.

A dead zone and repeat delay are used to reduce unintended changes when the sticks are near their center positions.

Move the sticks slowly and monitor the displayed values on the TFT LCD.

One-Burst (FC) Button L2 User Guide

I. Function Overview

This device supports two methods: automatic one-burst detection and manual one-burst marking. Manual marking can overwrite the automatic value and can also be undone and restored.

Operation Button Description

Manual One-Burst Marking L2 Marks FC at the current moment, overwriting the automatic detection value.

Undo Manual Marking R2 Restores the automatic value; if there is no automatic value, it completely clears the FC.

This is only effective during baking (PHASE_ROASTING) and curve recording; pressing L2 in other states will be ignored.

II. Button Operation Flow

1. L2 — Manual One-Burst Marking

When to Use: When you hear a one-burst sound, or determine that a one-burst has been reached, but the automatic detection has not yet marked it/the marking time is inaccurate.

Operation: Confirm that you are currently baking (ROAS status on the curve screen, time is running).

Press L2 once.

Upon success: The FC vertical line on the graph moves to the current time.

A yellow "M" appears next to the FC line, indicating manual marking.

The FC* flag appears at the bottom of the screen.

Repeatable press: Each press of L2 updates the FC time to the "current time," using the last press as the accurate time.

2. R2 — Cancel Manual Marking

When to use: When the manual marking position is incorrect and you want to restore it to the auto-detected value, or when you want to completely clear the FC.

Operation: Press R2 once.

Two possible results:

If FC was previously automatically detected (with a backup tAuto) → FC is restored to the auto-detected time, and the "M" disappears.

If FC was never automatically detected (only manually marked) → FC is completely cleared, and the FC line on the graph disappears.

Chapter 4. YEL and FC Temperature Threshold Adjustment

4.1 Definitions

Abbreviation	Full Name	Description
YEL	Yellowing	Yellowing-point temperature threshold
FC	First Crack	First-crack temperature threshold
NVS	Non-Volatile Storage	Persistent storage area

The YEL and FC thresholds are used as references for automatic event detection.

Default thresholds:

Parameter	Default Temperature
YEL	150°C
FC	183°C

4.2 Adjusting Thresholds with the PS2 Controller

Step 1: Enter Adjustment Mode

1. Confirm that the PS2 controller is connected.
2. Press and hold SELECT for at least 200 milliseconds.
3. The system enters YEL/FC threshold adjustment mode.
4. During this mode, the directional pad is used to adjust the temperature thresholds.

Step 2: Adjust the Temperatures

Controller Operation	Function	Adjustment
SELECT + ↑	Increase YEL	+1°C
SELECT + ↓	Decrease YEL	-1°C
SELECT + ←	Decrease FC	-1°C
SELECT + →	Increase FC	+1°C

The valid temperature range is greater than 50°C and less than 250°C. YEL must remain lower than FC.

Step 3: Save the Settings

1. Continue holding SELECT.
2. Press ○ Circle.
3. The system saves the thresholds to NVS.
4. Release SELECT after saving is complete.

The saved values are retained after a restart.

Step 4: Restore Default Thresholds

While in SELECT adjustment mode, press × Cross to restore:

- YEL = 150°C
- FC = 183°C

This operation resets and saves the threshold values. Do not confuse it with the normal bean-dropping function.

Chapter 5. BLE Configuration Using Serial Bluetooth Terminal

5.1 Preparation and Connection

1. Install Serial Bluetooth Terminal on an Android smartphone.
2. Power on the ESP32-S3.
3. Confirm that the firmware has started its BLE service.
4. Open Serial Bluetooth Terminal.
5. Open the device selection page and search for:

   `ESP32_Skycommand_BLE`
6. Connect to the device.
7. Open the terminal screen after the connection is established.

This firmware uses Bluetooth Low Energy (BLE), not the traditional Bluetooth SPP serial profile. Make sure the application and connection mode support BLE communication.

5.2 Command Input Rules

- Commands are case-insensitive.
- Separate parameters with semicolons (`;`).
- Send a newline character at the end of each command.
- Do not append units such as `°C` or `%` to numeric values.
- Back up the original recipe before making changes.

5.3 Viewing Recipes

To display all recipes:

```
RECIPESHOW;ALL
```

To display recipe 011:

```
RECIPESHOW;011
```

Supported recipe IDs:

Recipe ID	Description
011	Recipe 011
012	Recipe 012
013	Recipe 013
021	Recipe 021
022	Recipe 022
023	Recipe 023
031	Recipe 031
032	Recipe 032
033	Recipe 033

5.4 Recipe Modification Command Format

Use the `RECIPEDEF` command to modify a recipe.

General format:

```
RECIPEDEF;ID;ChargeTemp;ChargeHeat;ChargeAirflow;Stage1;Stage2;...;DropTemp
```

Each stage uses comma-separated values:

```
Temperature,Heat,Airflow
```

For example:

```
149,65,-1
```

This represents a stage at 149°C with heat set to 65 and airflow unspecified (`-1`).

The firmware supports up to six stages per recipe. Supply all required fields when modifying a recipe.

5.5 Example: Modify Recipe 011

The following command demonstrates the recipe format:

```
RECIPEDEF;011;200;65;65;111,70,-1;122,65,-1;132,60,70;149,65,-1;172,50,80;192
```

Field descriptions:

Field	Value	Description
Recipe ID	011	Recipe to modify
Charge temperature	200	Target charge temperature
Charge heat	65	Heat setting at charge
Charge airflow	65	Airflow setting at charge
Stage 1	111,70,-1	At 111°C, heat 70, airflow unspecified

Stage 2	122,65,-1	At 122°C, heat 65, airflow unspecified
Stage 3	132,60,70	At 132°C, heat 60, airflow 70
Stage 4	149,65,-1	At 149°C, heat 65, airflow unspecified
Stage 5	172,50,80	At 172°C, heat 50, airflow 80
Drop temperature	192	Target bean-drop temperature

These values are provided as a command-format example only. They are not a recommended roasting profile. Adjust the settings according to the bean variety, batch size, machine performance, and desired roast profile.

Important: After a successful `RECIPEDEF` command, the firmware automatically saves the recipe to NVS. Run `RECIPESHOW;011` again to verify the updated values.

5.6 Saving and Clearing Recipes

Command	Function
`RECIPESHOW;ALL`	Display all recipes
`RECIPESHOW;011`	Display recipe 011
`RECIPESAVE`	Manually save recipes
`RECIPECLEAR`	Clear stored recipe data in NVS

The `RECIPEDEF` command automatically saves the recipe after a successful update, so `RECIPESAVE` is normally unnecessary.

Warning: `RECIPECLEAR` removes stored recipe data. Do not use this command without first backing up your recipes. After clearing the stored data, the firmware may revert to its built-in default recipes.

5.7 Charge Temperature Limitations

The manual charge-temperature adjustment range is 160–200°C. However, recipe charge-temperature validation is determined by the firmware's `RECIPEDEF` implementation.

If an existing recipe contains a charge temperature outside the permitted range, the firmware may reject the modification. Query the recipe and check the applicable limits before making changes.

Chapter 6. Adjusting YEL/FC Through BLE

YEL and FC thresholds can also be modified through Serial Bluetooth Terminal.

6.1 Display Current Thresholds

`SHOWTHRESH`

The response displays the current YEL and FC temperatures and the relevant storage status.

6.2 Set YEL

Example: Set the yellowing threshold to 152°C.

```
SETYEL;152
```

6.3 Set FC

Example: Set the first-crack threshold to 185°C.

```
SETFC;185
```

6.4 Set Both Thresholds

```
SETTHRESH;152;185
```

The first value is YEL and the second value is FC.

6.5 Restore Default Thresholds

```
RESETTHRESH
```

After restoring the defaults, run:

```
SHOWTHRESH
```

to verify the resulting values.

Threshold Validation Rules

- YEL must be greater than 50°C and less than 250°C.
- FC must be greater than 50°C and less than 250°C.
- YEL must be lower than FC.
- Valid changes made through BLE are saved to NVS by the firmware.

Chapter 7. Operating Procedures

7.1 Preheating

1. Confirm that the controller, PS2 controller, and display are working properly.
2. Ensure that the roaster is in standby or a completed state.
3. Press △ Triangle to start preheating.
4. To cancel, press △ again during preheating or holding.

During roasting, △ has a different function: it toggles automatic bean dropping.

7.2 Charging Beans

1. Verify that the machine is ready for charging.
2. Press ○ Circle.
3. The system executes the charging procedure.

7.3 Adjusting Heat and Airflow

1. Use the right stick to adjust heat.
2. Use the left stick to adjust airflow.
3. Monitor the displayed values on the LCD.
4. Press R3 to set manual heat to zero.
5. Press L3 to set airflow to zero.

7.4 Dropping Beans

1. Confirm that the roaster is in the roasting stage.
2. Press × Cross.
3. The system executes the bean-dropping procedure.

Post-drop cleaning or cooling behavior depends on the firmware state machine and machine configuration.

7.5 Drum Control

Press START to toggle the drum between running and stopped. Verify that the mechanical system is safe before operating the drum.

7.6 Switching Display Modes

Press **L1** to switch between text and curve display modes.

SELECT is no longer used to switch displays. It is used for YEL/FC temperature threshold adjustment.

Chapter 8. Troubleshooting

Problem	Recommended Checks
PS2 controller does not respond	Check power, common ground, and DAT/CMD/ATT/CLK wiring
TFT LCD shows no image	Check power, SCLK, MOSI, RST, DC, and CS
Backlight is on but no text appears	Check ST7789 initialization, resolution, and GPIO settings

Stick values are unstable	Check stick centering, dead-zone settings, and signal quality
BLE connection fails	Confirm BLE service startup, device name, and connection mode
Commands receive no response	Check spelling, semicolons, newline settings, and BLE connection
Recipe modification is rejected	Check recipe ID, numeric limits, stage count, and field format
YEL/FC modification fails	Check temperature limits and ensure YEL is lower than FC

Chapter 9. Safety Precautions

1. Disconnect power before wiring or changing hardware connections.
2. Never connect a 5 V signal directly to an ESP32-S3 GPIO.
3. Verify machine conditions before operating heat, airflow, drum, or bean-drop functions.
4. Do not rely solely on the PS2 controller or TFT LCD for safety. Maintain appropriate hardware protection and emergency-stop mechanisms.
5. Back up existing recipes before making changes.
6. Do not use `RECIPECLEAR` without a backup.
7. Perform initial testing without activating the heater or mechanical loads.
8. YEL and FC are event-detection thresholds; they are not universal roasting standards for all bean varieties.

Chapter 10. Quick Reference

Operation	Method
Start preheating	△ Triangle
Charge beans	○ Circle
Drop beans	× Cross
Toggle automatic drop	□ Square; △ can also toggle it during roasting
Drum start/stop	START
Switch text/curve display	**L1**
Adjust heat	Right stick up/down
Adjust airflow	Left stick up/down
Set heat to zero	R3
Set airflow to zero	L3
Select recipe	↑ / ↓
Adjust charge temperature	← / →
Enter YEL/FC adjustment mode	Long-press SELECT
Increase/decrease YEL	SELECT + ↑ / ↓
Decrease/increase FC	SELECT + ← / →

Save YEL/FC	SELECT + ○
Restore default YEL/FC	SELECT + ×
Display all recipes	`RECIPESHOW;ALL`
Display recipe 011	`RECIPESHOW;011`
Modify a recipe	`RECIPEDEF;...`
Display YEL/FC thresholds	`SHOWTHRESH`
Set YEL	`SETYEL;152`
Set FC	`SETFC;185`
Set both thresholds	`SETTHRESH;152;185`
Restore default thresholds	`RESETTHRESH`

Chapter 11 Artisan HiBean as the Main Control

When using the Artisan HiBean as the main control, the PS2 controller can also be used to control firepower and ventilation. If the BLE connection is lost, the PS2 controller can continue to be used for control.

How to use: When the Artisan HiBean is preheated and you are ready to put in a bean, press ○ on the PS2 controller to put in the bean; the curve will be recorded simultaneously.